

Numerical Methods Assignment 1

Group XX

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Instructions:

- Submit the assignment by the deadline on September 16 at 14:00 via Brightspace. All group members must be enrolled in the group before the assignment is submitted.
- Submit two files: this .mlx file and an .pdf export of the same file. Save both files as NM_AS1_GroupX.mlx and NM_AS1_GroupX.pdf, replacing the X by your group number.
- Ensure your group number, names and student numbers are at the top of the document.
- **Consult the course description for details** on the assignment, including information on the discussion and the generative AI policy.
- Include a statement on the ways in which you used generative AI, if any, in the box below. If you worked individually, ensure every group members writes their own statement.
- This type of MATLAB file is called a Live Code File Format (.mlx). It allows you to have all code in one place, and show code, generated figures, text, and equations all next to each other. It is fairly self-explanatory, but this page and links within it provide some further explanation: http://nl.mathworks.com/help/matlab/matlab_prog/what-is-a-live-script-or-function.html. You are welcome to remove the text of the assignment or keep it and add in your solutions.

Generative AI statement

Explain here how you used generative AI, if any, for this assignment. If you worked individually, use a separate statement for each group member.

Set-up (12 pts.)

This assignment is on the root-finding methods that are discussed in Chapter 4 of the lecture notes. As you will require an implementation for each of the four methods, start by defining these in the next few blocks of code. The first three tutorials provide instructions on how to implement these functions. Use the stopping criteria suggested during lectures.

Ensure you are calling the functions you define here in Problems 1-3.

```
% Implementation of the bisection method
function [root, rootvec] = bisection_method(fun, l, u, maxit, tol)

end
```

% Implementation of fixed-point iteration

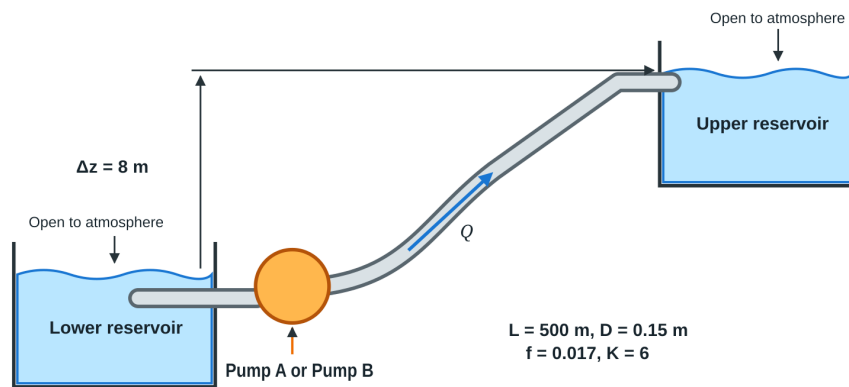
% Implementation of Newton-Raphson

% Implementation of the secant method

Problem 1: Pump operating point and energy consumption (30 pts.)

Problem description

A pumping system transports water from a lower reservoir to an upper reservoir located 8 m higher. The reservoirs are connected by a 500 m-long pipe with an internal diameter of 0.15 m. See below a diagram of the situation:



Two pumps are available: pump A and pump B. In this question, you will decide whether you prefer pump A or pump B. We start off by providing all the information necessary for you to answer this question.

Pump characteristic curves

The pressure head supplied by each pump H_{pump} (expressed in m) is represented by the following empirical equation:

$$H_{\text{pump}}(Q) = H_0 - bQ - aQ^2 - cQ^3,$$

where Q is the volumetric flow rate, expressed in m^3/s , H_0 is the maximum pressure head produced by the pump when the flow rate is zero. This value is also called the shut-off head. The coefficients a , b , and c describe how the pump head decreases as the flow rate increases:

- b controls the initial linear decrease in the pump head.
- a represents the quadratic curvature of the pump characteristic curve.
- c captures the stronger nonlinear reduction in pump head at high flow rates.

These coefficients are generally obtained by fitting the polynomial model to experimental pump-performance data supplied by the manufacturer. The supplier has supplied the following parameters.

Pump parameters

Parameter	Pump A	Pump B	Units
H_0	34	42	m
b	100	80	$\frac{s}{m^2}$
a	8,000	12,000	$\frac{s^2}{m^3}$
c	50,000	30,000	$\frac{s^3}{m^6}$
Efficiency η	0.70	0.82	

System curve

The pressure head required by the piping system H_{system} (expressed in m) is given by

$$H_{\text{system}}(Q) = \Delta z + \left(f \frac{L}{D} + K \right) \frac{v(Q)^2}{2g},$$

where Q is again the flow rate. The mean velocity $v(Q)$ is related to the flow rate Q according to the equation

$$v(Q) = \frac{Q}{A},$$

where A is the cross-sectional area of the pipe. The system described use the following system parameters

Parameter	Symbol	Value
Pipe length	L	500 m
Pipe diameter	D	0.15 m
Elevation difference	Δz	8 m
Darcy friction factor	f	0.017
Minor-loss coefficient	K	6
Gravitational acceleration	g	9.81 m/s^2

Assume that the friction factor remains constant over the operating range.

Operating point

The operating point Q^* occurs when the pressure head supplied by the pump is equal to the pressure head required by the piping system:

$$H_{\text{pump}}(Q^*) = H_{\text{system}}(Q^*)$$

Because the pressure head supplied by a pump depends on the flow rate, the operating flow rate is not known in advance.

Energy-consumption analysis

We can calculate the electrical energy consumed during one hour of continuous operation of a pump.

Hydraulic power

The hydraulic power (in kilowatts) transferred from the pump to the water is

$$P_{\text{hydraulic}} = \frac{\rho g Q H}{1000},$$

where:

- Q is the operating flow rate in m^3/s .
- H is the head at the operating point in m.
- $P_{\text{hydraulic}}$ is the hydraulic power in kW.
- ρ is the water density, use $\rho = 998 \text{ kg/m}^3$.

Electrical power

The electrical power required by the pump is

$$P_{\text{electrical}} = \frac{P_{\text{hydraulic}}}{\eta},$$

where η is the efficiency of the pump.

Electrical energy consumed per hour

Electrical energy is calculated as

$$E = P_{\text{electrical}} \cdot t.$$

When the electrical power is expressed in kW and the operating time is expressed in hours, the resulting electrical energy is expressed in kWh.

Problem 1:

Considering the information above and the problem description,

- Compute the operation point for both pumps. Use all four numerical methods for root-finding discussed and generate convergence plots. Explain what choices you made for each method and motivate them. Clearly

explain the pros and cons of each method, based on the observed behaviour, and motivate a decision for the best numerical method. **(15 pts.)**

b. Choose one of the pumps based on its power consumption and the volume it can transport. **(5 pts.)**

Note: All of your calculations and reasoning must be clearly explained as text, in order to receive full marks.

Solutions problems 1a and 1b

Add your solution here using written text, equations, and blocks of code.

Problem 1c: challenging extension on flow-dependent friction factor (10 points)

In Problem 1a-b, the friction factor was assumed to be constant, $f = 0.017$. A more realistic model calculates f using the Colebrook equation when the flow is turbulent, as we have seen in the lectures. The internal roughness of the pipes are slightly different. The roughness of pump A is $k_A = 4.5 \times 10^{-5}$ m and of pump B $k_B = 3 \times 10^{-5}$ m. Use for the dynamic viscosity of water $\mu = 10^{-3}$ kg/(ms).

Note that the Colebrook equation is only valid for flows with Reynolds number greater than 4000. For all smaller Reynolds numbers, extend the laminar friction factor equation:

$$f = \frac{64}{Re}.$$

Determine a new operating point and choose the pipe with the lowest energy consumption. You need not compare all numerical methods, but ensure that you motivate the choices that you made. Clearly explain your full approach.

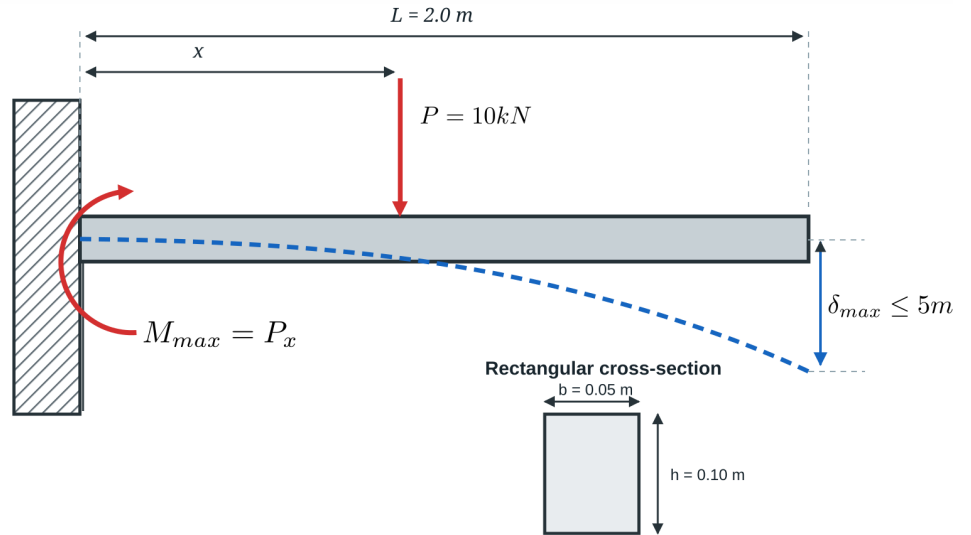
Solution problem 1c

Add your solution here using written text, equations, and blocks of code.

Problem 2: Cantilever Beam (26 pts.)

Problem description

A machine with a weight of $P = 10$ kN must be installed on a cantilever beam. The beam is fixed at its left end and has a total length of $L = 2.0$ m. The machine is modeled as a concentrated vertical load acting at a



distance x from the fixed support.

The position of the machine must satisfy three design requirements:

- The maximum bending stress at the fixed support must not exceed the allowable stress of the material.
- The maximum vertical deflection of the beam must not exceed the allowable deflection.
- The load position must, naturally, satisfy:

$$0 \leq x \leq L.$$

We provide further information on the properties of the beam and how to calculate maximum bending stress and maximum vertical deflection below. The questions of the problem are stated after that.

Beam and material properties

The beam is made of steel and has a rectangular cross-section. Use the following properties:

Parameter	Symbol	Value
Beam length	L	2.0 m
Applied load	P	10 kN
Young's modulus	E	200 GPa
Cross-section width	b	0.05 m
Cross-section height	h	0.10 m
Allowable bending stress	σ_{allow}	150 MPa
Allowable deflection	δ_{allow}	5 mm

Cross-section properties

For a rectangular cross-section, the second moment of area about the neutral axis is

$$I = \frac{bh^3}{12}$$

The distance from the neutral axis to the outermost fibre is

$$c = \frac{h}{2}$$

Stress constraint

For a concentrated load P applied at a distance x from the fixed support, the maximum bending moment occurs at the fixed support:

$$M_{\max} = Px.$$

The corresponding maximum bending stress is

$$\sigma_{\max}(x) = \frac{M_{\max}c}{I}.$$

The beam satisfies the stress requirement when

$$\sigma_{\max}(x) \leq \sigma_{\text{allow}}$$

The maximum allowable position according to the stress constraint is denoted by $x_{\max}$.

Deflection constraint

The load is not necessarily applied at the free end of the beam. The beam continues from the load position x to the free end at L . The vertical deflection at the load position is

$$\delta_P(x) = \frac{Px^3}{3EI}$$

The free-end deflection is

$$\delta_{\text{tip}}(x) = \frac{Px^2(3L - x)}{6EI}$$

The beam satisfies the deflection requirement when

$$\delta_{\max}(x) \leq \delta_{\text{allow}},$$

where $\delta_{\max}$ is the maximum deflection, which happens either at the load position or at the free-end.

Safety factor

Define the safety factor against bending as

$$n = \frac{\sigma_{\text{allow}}}{\sigma_{\max}}$$

Problem 2:

Complete the following tasks:

- Determine the maximum distance $x_{\max}$ from the fixed support at which the force can be applied while satisfying all design requirements. Use root-finding methods and explain which method performs best. As

before, explain in detail how the methods were used and generate convergence plots. Describe pros and cons of each method to motivate the best choice. **(16 pts.)**

b. Identify which design requirement governs the maximum allowable position: beam length, allowable bending stress, or allowable deflection. Explain the physical meaning of the governing requirement. **(5 pts.)**

c. Once you have identified the maximum position of the load, determine the safety factor. If it is less than 2, identify which cross-sectional area (width or height) should be modified and what its value should be to meet this requirement. **(5 pts.)**

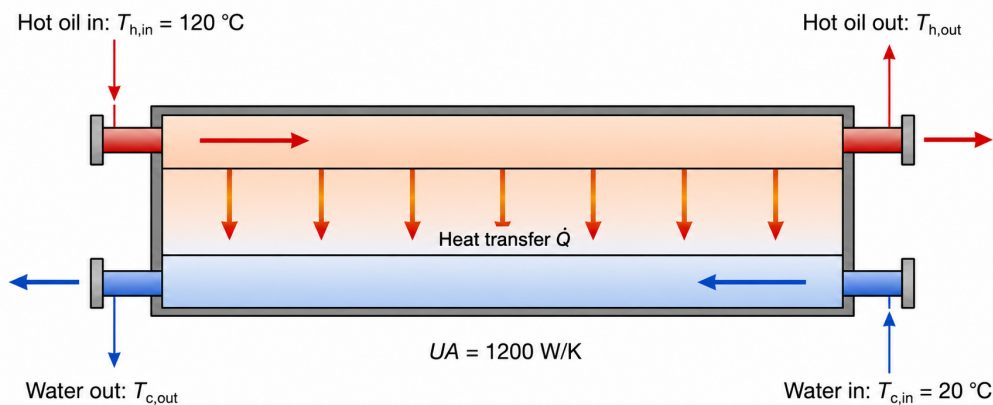
Solution problem 2

Add your solution here using written text, equations, and blocks of code.

Problem 3: Outlet Temperatures of a Counter-Flow Heat Exchanger (logarithmic mean temperature difference -LMTD) **(22 pts.)**

Problem description

A counter-flow heat exchanger is used to heat water using hot oil. The two fluids flow in opposite directions and do not mix (see <https://www.youtube.com/watch?v=NvkZaWLe0Sk> for an explanation of the device). A simple counter-flow heat exchanger is shown in the below figure:



The inlet temperatures, mass flow rates, specific heat capacities, and overall thermal conductance of the heat exchanger are known. However, the outlet temperatures and the heat-transfer rate are unknown.

The goal of this problem is to determine **(6 pts)**:

1. The outlet temperature of the water
2. The outlet temperature of the oil
3. The heat-transfer rate between the fluids

Explain the process and methodology, developing the energy balance, applying all the root-finding methods studied in lectures, and comparing their convergence rates and errors to assess robustness. Again, be sure to motivate and lay-out clearly all decisions you make **(16 pts.)**

The necessary information to solve this problem is below.

Problem variables

Use the following input values for the particular case:

Quantity	Symbol	Value	Units
Oil mass flow rate	$\dot{m}_o$	2.0	kg/s
Water mass flow rate	$\dot{m}_w$	1.0	kg/s
Oil specific heat capacity	$c_{p,o}$	2000	J/(kg K)
Water specific heat capacity	$c_{p,w}$	4180	J/(kg K)
Oil inlet temperature	$T_{o,in}$	120	°C
Water inlet temperature	$T_{w,in}$	20	°C
Overall thermal conductance	UA	1200	W/K

The heat capacities are representative constant values for liquid water and a typical heat-transfer oil.

The subscripts have the following meanings:

- o : oil
- w : water
- in : inlet
- out : outlet

Modeling assumptions

Assume that:

1. The heat exchanger operates under steady-state conditions
2. There is no heat loss to the surroundings
3. The two fluids do not mix
4. The specific heat capacities remain constant
5. There is no phase change
6. Kinetic- and potential-energy changes are negligible
7. The overall heat-transfer coefficient is constant
8. The heat exchanger operates in counter-flow

Because there is no heat loss to the surroundings, the heat lost by the oil is equal to the heat gained by the water.

Heat-capacity rates

For each fluid, define the heat-capacity rate as

$$C = \dot{m}c_p$$

For the oil,

$$C_o = \dot{m}_o c_{p,o}$$

and for the water,

$$C_w = \dot{m}_w c_{p,w}$$

The units of a heat-capacity rate are

$$[C] = \frac{\text{kg}}{\text{s}} \frac{\text{J}}{\text{kg K}} = \frac{\text{W}}{\text{K}}$$

The heat-capacity rate indicates the amount of heat-transfer power required to change the fluid temperature by one kelvin.

Mathematical setup

The solution must satisfy the energy balance

$$\dot{Q} = C_w(T_{w,\text{out}} - T_{w,\text{in}}) = C_o(T_{o,\text{in}} - T_{o,\text{out}}),$$

where $\dot{Q}$ is the heat-transfer rate.

For counter-flow operation, the terminal temperature differences are

$$\Delta T_1 = T_{o,\text{in}} - T_{w,\text{out}}, \quad \Delta T_2 = T_{o,\text{out}} - T_{w,\text{in}}$$

The logarithmic mean temperature difference and exchanger heat-transfer equation are

$$\Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)}, \quad \dot{Q} = UA\Delta T_{\text{lm}}$$

These equations contain the two unknowns $T_{w,\text{out}}$ and $T_{o,\text{out}}$. To be able to determine them both, first rewrite the equations so that there exists a single equation with one unknown. Then apply a root-finding method.

Solution problem 3

Add your solution here using written text, equations, and blocks of code.